ADITYA JADHAV

Backend Engineer | Production AI Systems (LLM, RAG, NL-to-SQL) | Python, FastAPI, PostgreSQL, AWS

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SUMMARY

Backend engineer who shipped production AI systems: an NL-to-SQL copilot whose self-healing pipeline cut failed queries by 85%, and a real-time proctoring platform that runs fully offline. Led a 5-person team across three production systems at EduDiagno (Jan–Jun 2026). Strongest in PostgreSQL, FastAPI, RAG pipelines, and AWS deployment.

EXPERIENCE

Backend Engineer & Team Lead Jan 2026 – Jun 2026
EduDiagno Pvt. Ltd. *Bengaluru (Remote)*

- Appointed team lead in the first week of a 5-engineer AI hiring platform initiative; owned technical direction and delivery across three production systems.
- Cut failed NL-to-SQL queries by 85% by building a self-healing pipeline that classifies the error, re-prompts with the failure attached, and retries. Built the Vanna-based BI copilot with RAG schema retrieval (Qdrant, schema-only embeddings) across PostgreSQL, MySQL, MSSQL, and Redshift.
- Enforced safe analytics access with 4-level RBAC, read-only SQL validation, guardrails against destructive queries, and encrypted per-database credentials.
- Delivered a multi-tenant exam platform from a one-line brief: OTP auth, request-based enrollment, scheduling, a 15+ model normalized PostgreSQL schema, and offline/LAN deployment.
- Built a standalone Socket.IO proctoring server (tab-switch, copy-paste, multi-login detection), decoupled from the main app for independent scaling.
- Built Parakeet, an AI interview copilot with real-time voice transcription and live answer suggestions (OpenAI, Deepgram).
- Migrated infrastructure from on-premise to AWS (EC2, RDS, VPC); containerized services with Docker; set up CI/CD with GitHub Actions through a bastion host, with pre-migration pg_dump backups.

PROJECTS

Voice RAG System | *Python, FastAPI, Qdrant, Redis, PostgreSQL, Next.js/TypeScript* HH Goa Hackathon, 2026

- Hybrid dense + sparse retrieval with RRF fusion and cross-encoder reranking, plus four chunking strategies.
- Redis-backed semantic cache, three-tier guardrails, and a custom orchestrator (no LangChain/LlamaIndex) with per-stage latency timing; Sarvam AI speech-to-text and Anthropic API.

Active Products — E-Commerce Platform | *Next.js, TypeScript, Prisma, PostgreSQL (Neon), Razorpay* activeproducts.in

- Live store with trader wholesale flow, admin dashboard, PDF receipts, and reviews; Razorpay signature-verified webhooks with a reconciliation job for stuck payments; Upstash Redis rate limiting.
- Found and fixed two IDOR vulnerabilities; wrote India-compliant legal pages (DPDP Act 2023, Consumer Protection Act 2019, FSSAI).

TECHNICAL SKILLS

Languages	Python, TypeScript/JavaScript, SQL
Backend	FastAPI, Node.js/Express, Django, Socket.IO/WebSockets, Prisma
AI / LLM	RAG, hybrid retrieval and reranking, NL-to-SQL, Vanna, Qdrant, LangChain, OpenAI, Anthropic API, Deepgram
Data	PostgreSQL, MySQL, MSSQL, Redshift, Redis
Cloud / DevOps	AWS (EC2, RDS, VPC), Docker, GitHub Actions, CI/CD, Vercel
Security	RBAC, JWT/OTP auth, rate limiting, CORS, SQL safety validation
Frontend	Next.js, React, Tailwind CSS

ACHIEVEMENTS

- **National Hackathon Runner-Up**, YCCE Nagpur: Rust-based high-frequency trading bot for liquidity management.
- **ISRO ASCEND 2026**: selected for India's aerospace and autonomous drone engineering challenge.
- **Appreciation letter**, EduDiagno: recognition for independent backend architecture and production contributions.
- **Student Head, SIH**: coordinated 600+ participants. Technical lead for department events and college cultural fest.

EDUCATION

B.Tech, Computer Science Engineering 2022 – 2026
MIT Chhatrapati Sambhajinagar CGPA: 8.54